DATA 211– Introduction to Data Science

LECTURE 1

Prof Dr Usman Qamar

Prof Usman Qamar

Joined KFUPM in 2025

- Professor of Data Sciences & GenAI, Maths department and Information and Computer Science Department, KFUPM
- SDAIA-KFUPM Joint Research Center for Artificial Intelligence

NUST, Pakistan in 2011-225

- Tenured Professor of Data Sciences & GenAI, Computer & Software Engineering Department, College of E&ME
- xHoD Computer & Software Engineering Department, NUST (2019-2024)

Previous Experience

- Office of National Statics, London, UK
- Vodafone, UK

Prof Usman Qamar

Education

- Post-Doc, University of Manchester, UK
- PhD, University of Manchester, UK, 2008
- MPhil, University of Manchester, UK, 2005
- MS, University of Manchester Institute of Science and Technology (UMIST), UK, 2003
- B.E, National University of Sciences and Technology (NUST), Pakistan, 2001

Prof Usman Qamar

Fellow

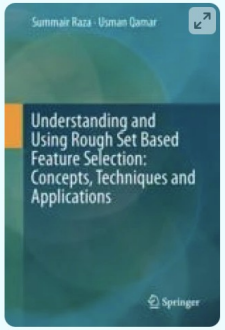
- The Engineering and Physical Sciences Research Council-UK Fellow
- Charles Wallace Fellow
- British Council Fellow







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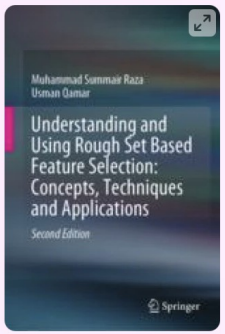
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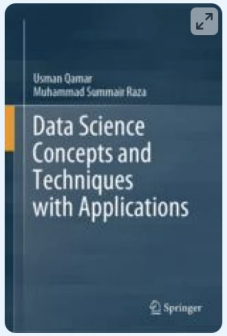
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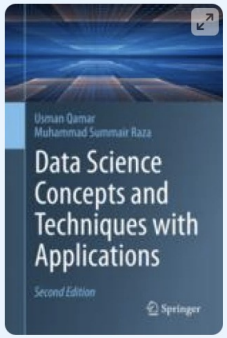
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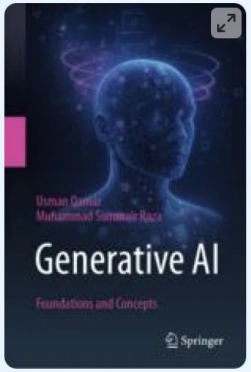
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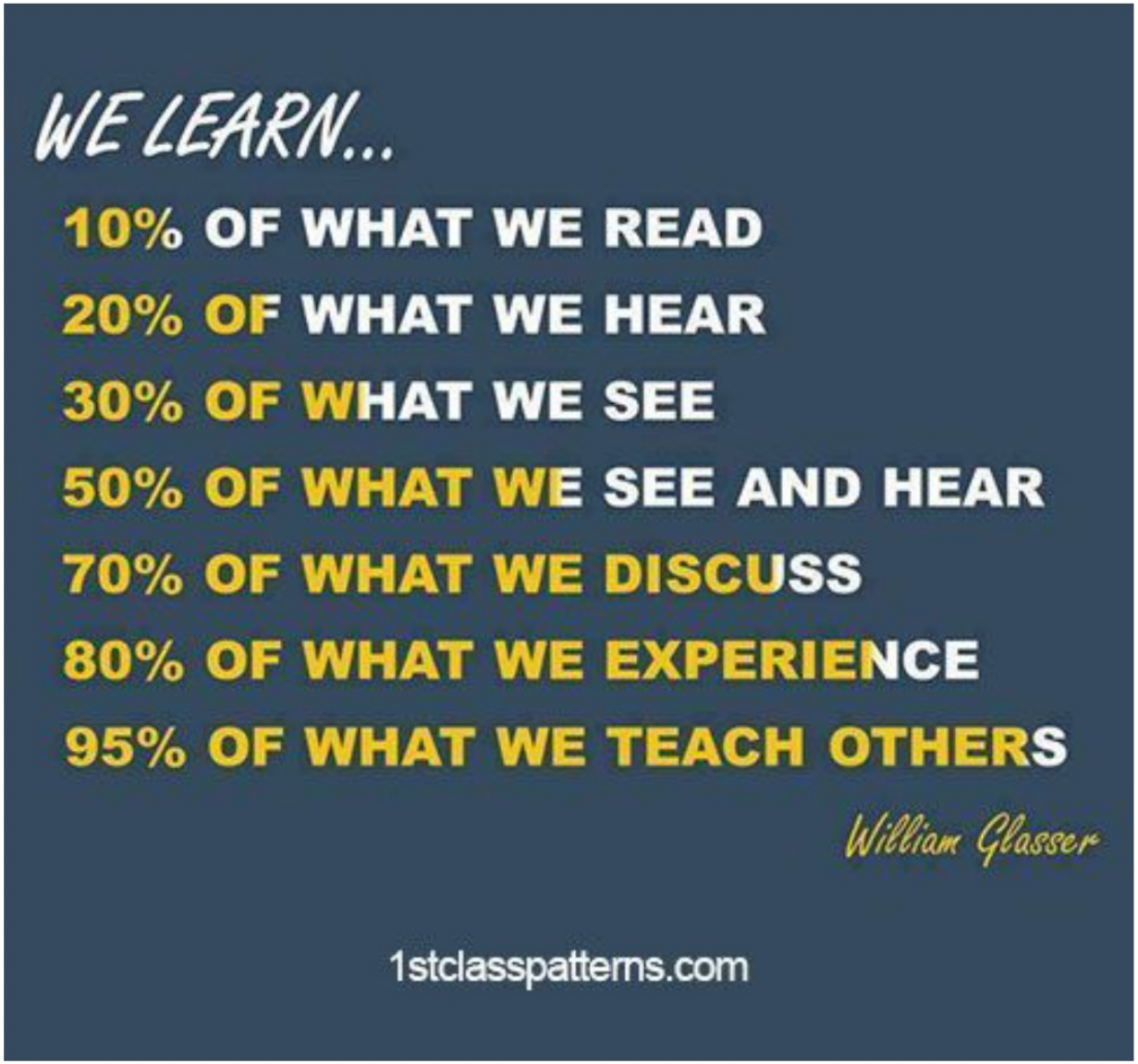


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Foundations & Concepts

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WE LEARN...
10% OF WHAT WE READ
20% OF WHAT WE HEAR
30% OF WHAT WE SEE
50% OF WHAT WE SEE AND HEAR
70% OF WHAT WE DISCUSS
80% OF WHAT WE EXPERIENCE
95% OF WHAT WE TEACH OTHERS
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AGENDA

What is Data Science?

Course Introduction

Course Details

Big Data Everywhere!

BIG DATA

Data that is TOO LARGE & TOO COMPLEX for conventional data tools to capture, store and analyze.

Shares traded on US Stock Markets each day:

7 Billion

Data generated in one flight from NY to London:

10 Terabytes

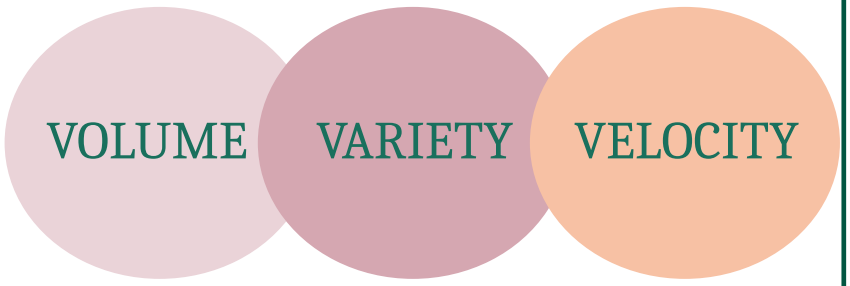
Number of tweets per day on Twitter:

400 Million

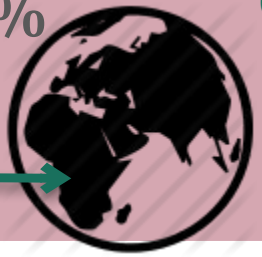
Number of 'Likes' each day on Facebook:

3 Billion

The 3V's of Big Data



90% OF THE WORLD'S DATA WAS GENERATED IN THE **LAST TWO YEARS**





Data on its own is useless unless you can make sense of it!

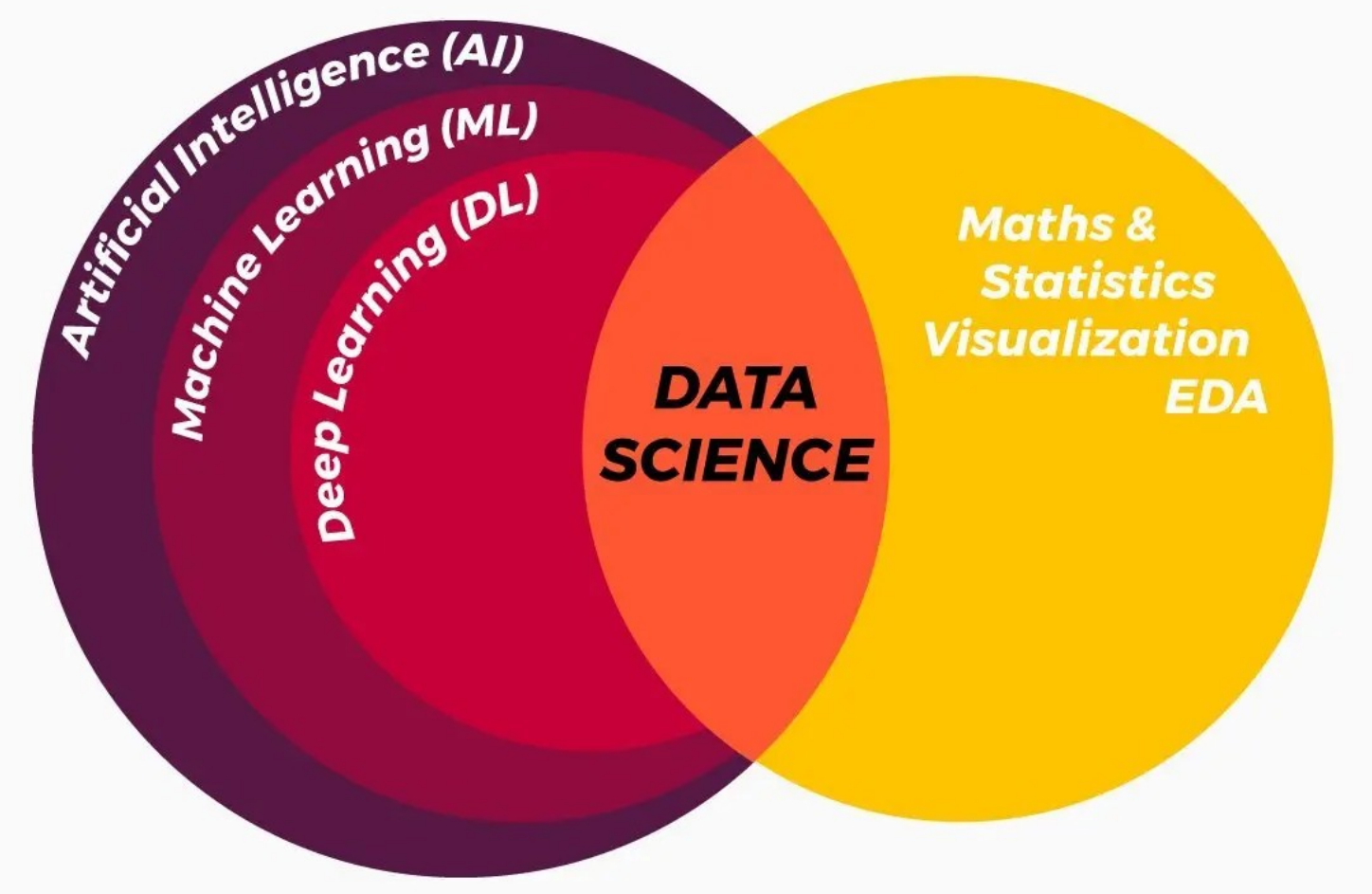
WHAT IS DATA SCIENCE?

Preface

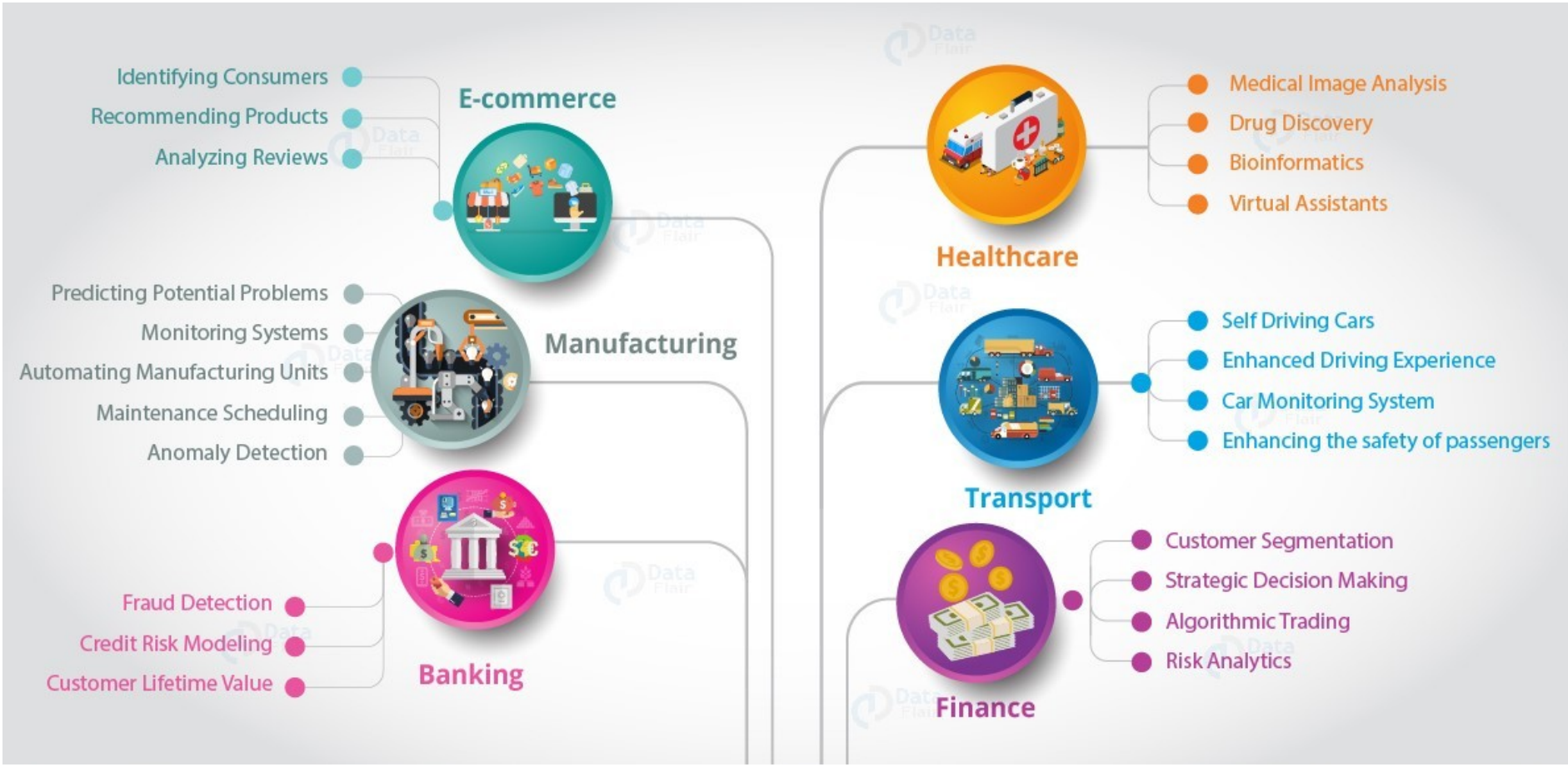
As this book is about data science, the first question it immediately begs is: What is *data science*? It is a surprisingly hard definition to nail down. However, for us, data science is perhaps the best label for the cross-disciplinary set of skills that are becoming increasingly important in many applications across industry and academia. It comprises three distinct and overlapping areas: firstly, *statistician* who knows how to model and summarize the data, *data scientist* who can design and use algorithms to efficiently process and visualize the data, and finally the *domain expert* who will formulate the right questions and put the answers in context. With this in mind, we would encourage all to think of data science not as a new domain of knowledge to learn, but a new set of skills that you can apply within your current area of expertise.

The book is divided into three parts. The first part consists of the first three chapters. In Chap. 1, we will discuss the data analytics process. Starting from the basic concepts, we will highlight the types of data, its use, its importance, and issues that are normally faced in data analytics. Efforts have been made to present the concepts in the most simple possible way, as conceptual clarity before studying the




Artificial Intelligence (AI)
Machine Learning (ML)
Deep Learning (DL)
DATA SCIENCE
Maths &
Statistics
Visualization
EDA

DATA SCIENCE APPLICATION EXAMPLES



DATA SCIENCE WORKFLOW

WHAT IS DATA SCIENCE?

01
BUSINESS UNDERSTANDING

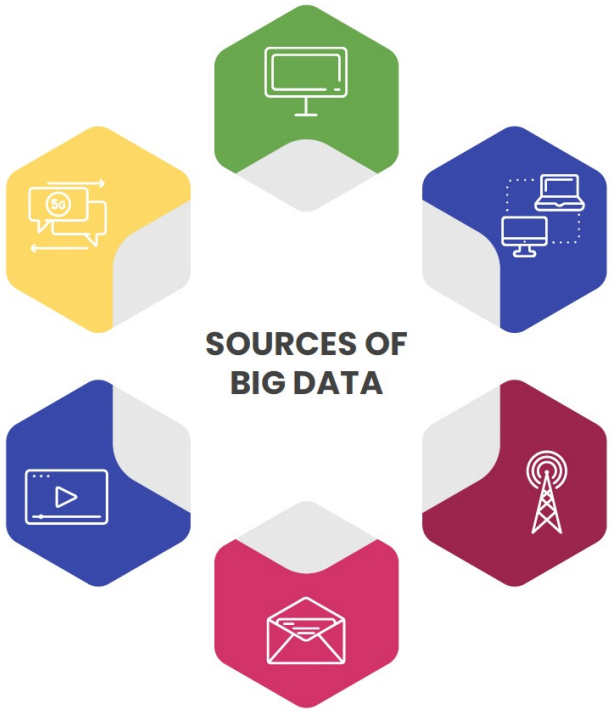
Identify the problem that must be considered in the study

02
DATA COLLECTION

Collect data that serves your study's objectives

03
DATA CLEANING

Fix the consistency in the data and handle missing values



04
FEATURE ENGINEERING

Transform your raw data into relevant and meaningful features

05
PREDICTIVE MODELS

Build Models
Train machine/deep learning models, and evaluate their performance and use them to make predictions

06
DATA VISUALIZATION

Communicate the finding with stakeholders and illustrate them with interactive visualization

Phase 1 – Problem Understanding

Objective Clarity

The primary objective is to solve the right problem, not just any problem. Misalignment between business and analytics teams can lead to ineffective solutions.



Key Steps

- Define Project Objectives: Clearly articulate what the business aims to achieve (e.g., increase customer retention).
- Translate to Data historical data).

Science: Convert business objectives into a measurable data science problem (e.g., predict customer churn using

Tip: Always revisit this phase after model evaluation to ensure the solution truly addresses the initial business challenge.

Phase 2 – Data Preparation

This is the most time-consuming phase, often consuming 60-80% of project time, as raw data is messy and requires significant cleaning and reshaping for modeling.

Outlier Handling

Identify and address anomalous data points that can skew results.

Standardization & Normalization

Scale variables to a consistent range, crucial for many machine learning algorithms.

Reclassification & Binning

Group categorical or numerical data into meaningful categories or bins.

Indexing & Unique IDs

Ensure each record has a unique identifier for proper data management.

Tools: Python (pandas), R (dplyr, tidyverse) are essential for efficient data manipulation.

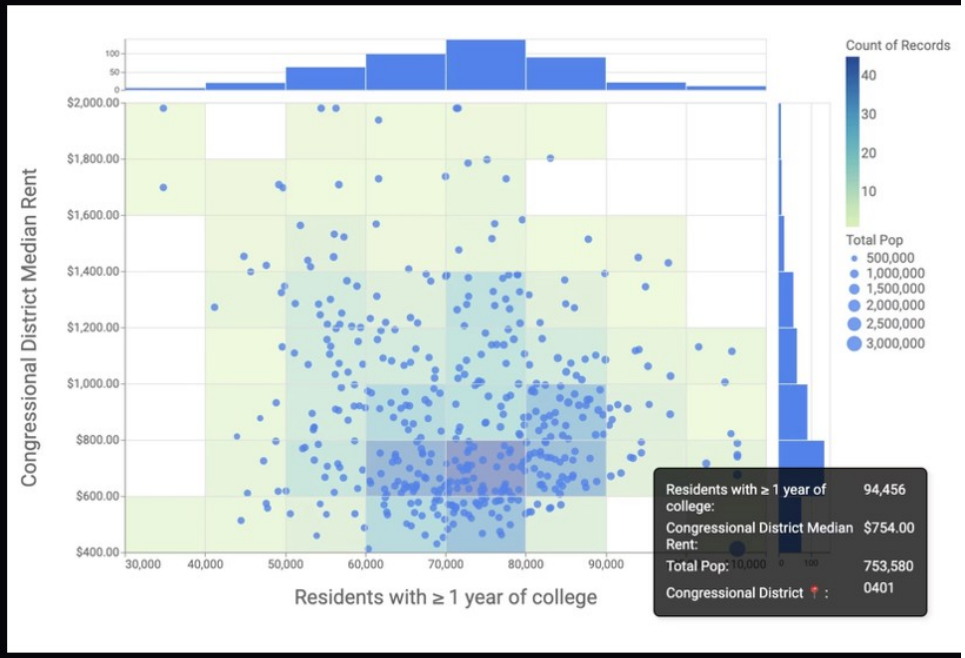
Exploratory Data Analysis (EDA)

EDA is critical for understanding data structure and patterns before modeling, primarily through visualization and descriptive statistics.

Analytical Focus

- **Univariate analysis: Histograms and boxplots reveal single variable distributions.**
- **Bivariate analysis: Scatterplots and correlation heatmaps show relationships between two variables.**
- **Multivariate analysis: Uncover complex patterns across multiple variables.**

Tools: Python (matplotlib, seaborn) and R (ggplot2) are vital for effective visualizations.



Phase 5: Modeling

This phase focuses on applying advanced algorithms to extract valuable insights and patterns from prepared data.

Algorithm Selection

Choose suitable algorithms (e.g., regression, decision trees, neural networks) based on data type and problem.

Baseline Performance

Ensure new models significantly outperform simple baseline predictions.

Hyperparameter Tuning

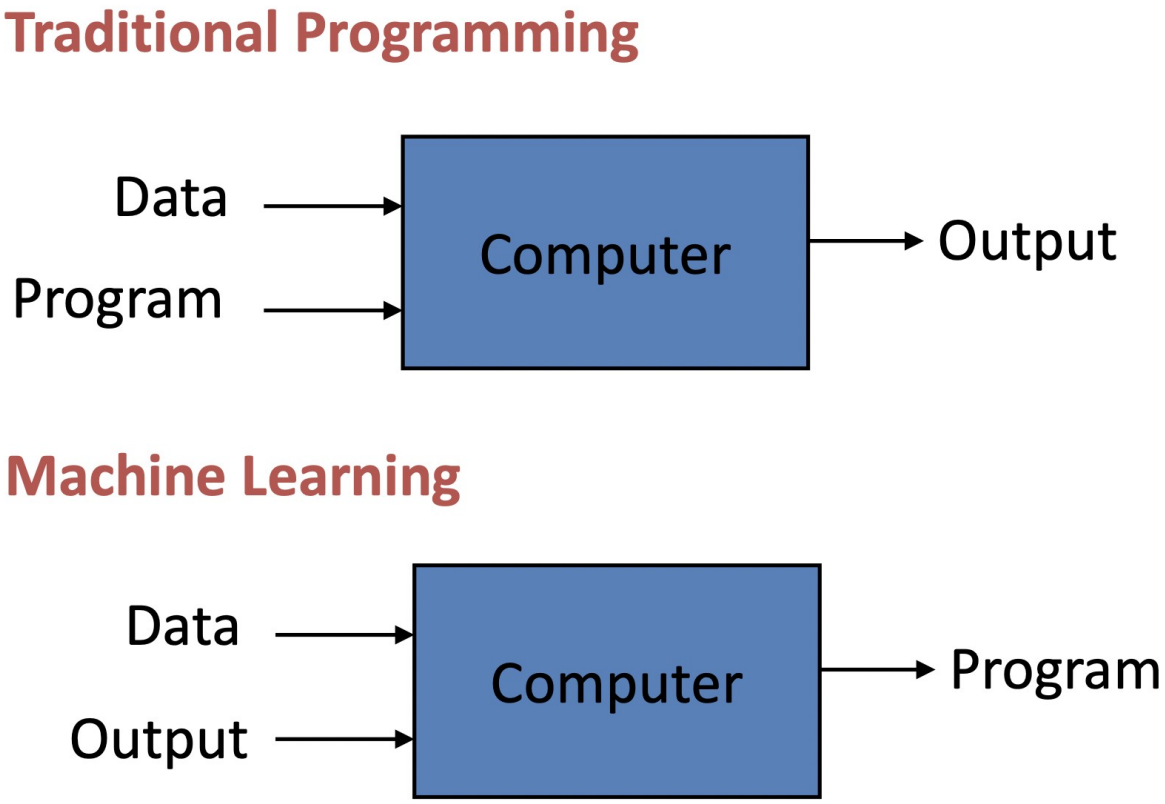
Optimize model performance through systematic techniques like grid or random search.

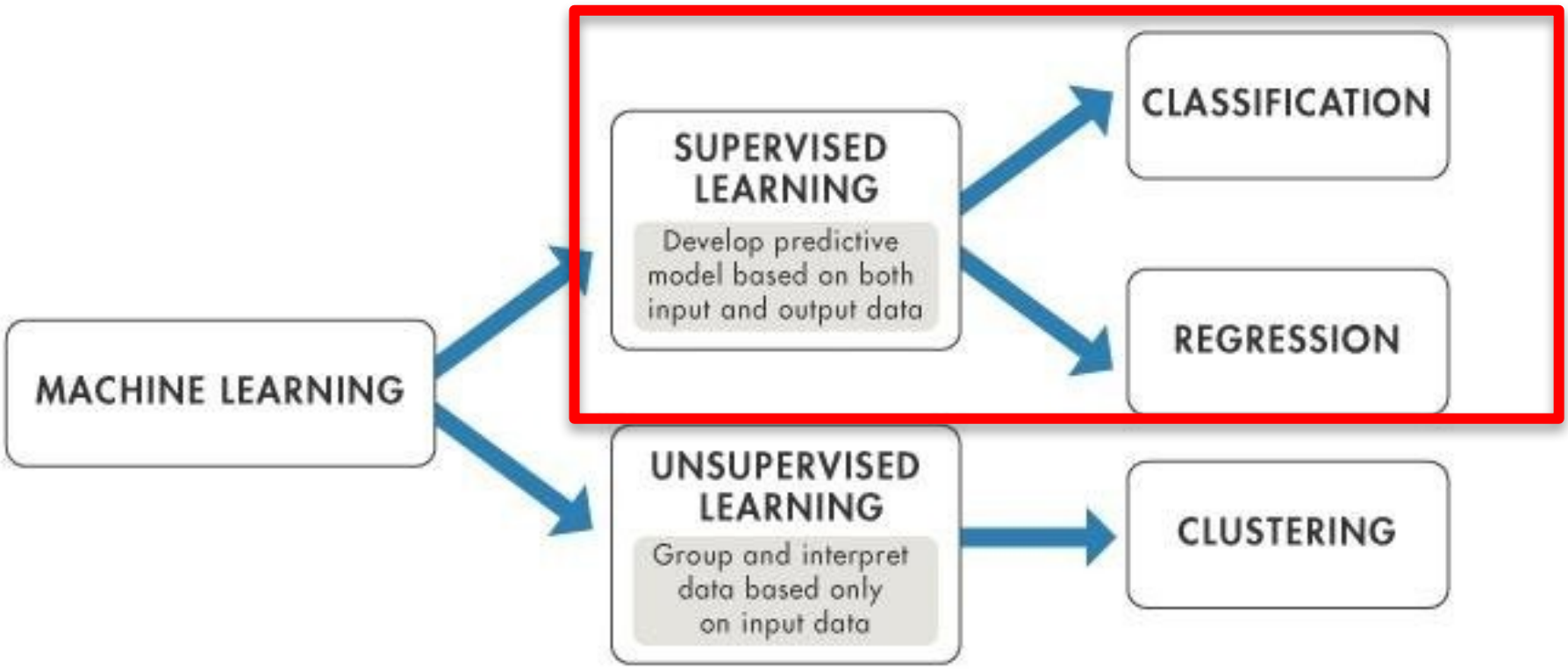
Cut-off Optimization

Select optimal decision thresholds to balance objectives, such as precision vs. recall.

It's crucial to align model selection and optimization with specific business goals, like minimizing false negatives in loan default predictions.

- Machine learning approach: program an algorithm to automatically learn from previous data.




MACHINE LEARNING
SUPERVISED LEARNING
Develop predictive model based on both input and output data
CLASSIFICATION
REGRESSION
UNSUPERVISED LEARNING
Group and interpret data based only on input data
CLUSTERING

Phase 6: Evaluation – Assessing Model Utility

A technically "good" model isn't always useful. Evaluation ensures the model addresses the original business problem and its real-world implications.

Key Metrics

- Classification: Accuracy, F1-Score, ROC AU
- Regression: RMSE, MAE, R^2

Critical Questions

- Does it beat the baseline?
- Does it solve the original business proble
- Are error costs acceptable?
- Are results generalizable to new data?



Essential Tools for Data Science

A robust toolkit is fundamental for any data scientist. These tools span programming languages, specialized libraries, development environments, and database management systems, enabling efficient data manipulation, analysis, and model deployment.

Programming Languages

Python and R are the cornerstones for statistical analysis, machine learning, and data visualization due to their extensive libraries and community support.

Core Libraries

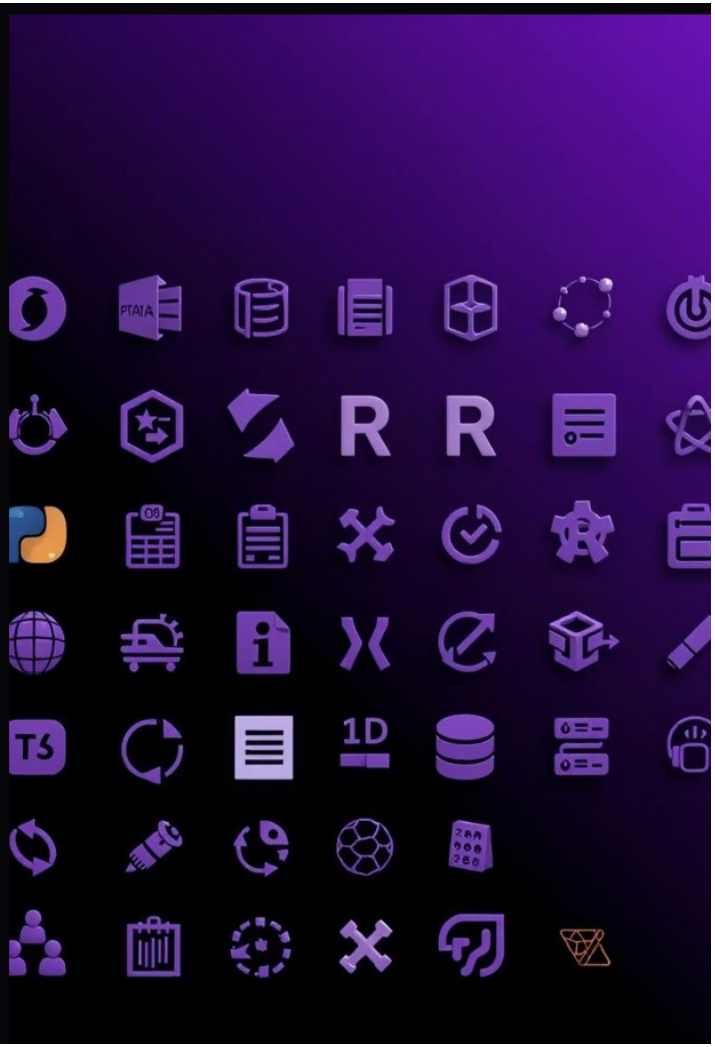
NumPy for numerical operations, Pandas for data manipulation, Scikit-learn for machine learning, and ggplot2 for advanced visualizations.

Development Environments

Jupyter Notebook and RStudio provide interactive platforms for coding, experimentation, and sharing data science projects.

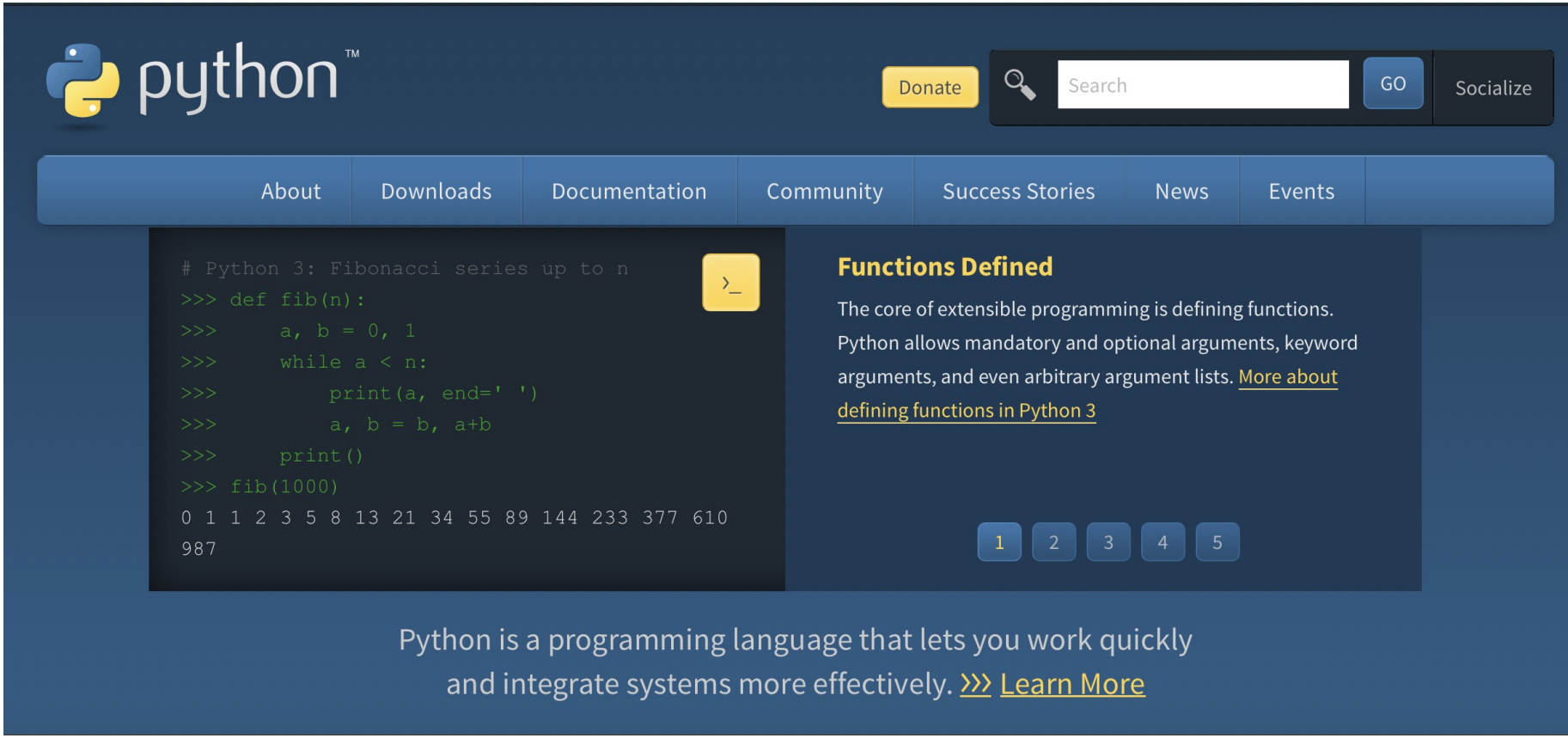
Database Systems

SQL for relational databases and MongoDB for NoSQL databases are essential for managing and querying large datasets.



COURSE CONTENT

- Basic statistics: Variance, Covariance, Correlation, Confidence Interval, Histogram
- Data frames, series, slicing, sorting
- Data acquisition, cleaning, manipulation
- Analytical methods: PCA, Linear Regression, Gradient Descent, Bayesian Classifier, Decision Tree
- Clustering: K-means, Hierarchical
- Big data concepts: MapReduce, Hadoop

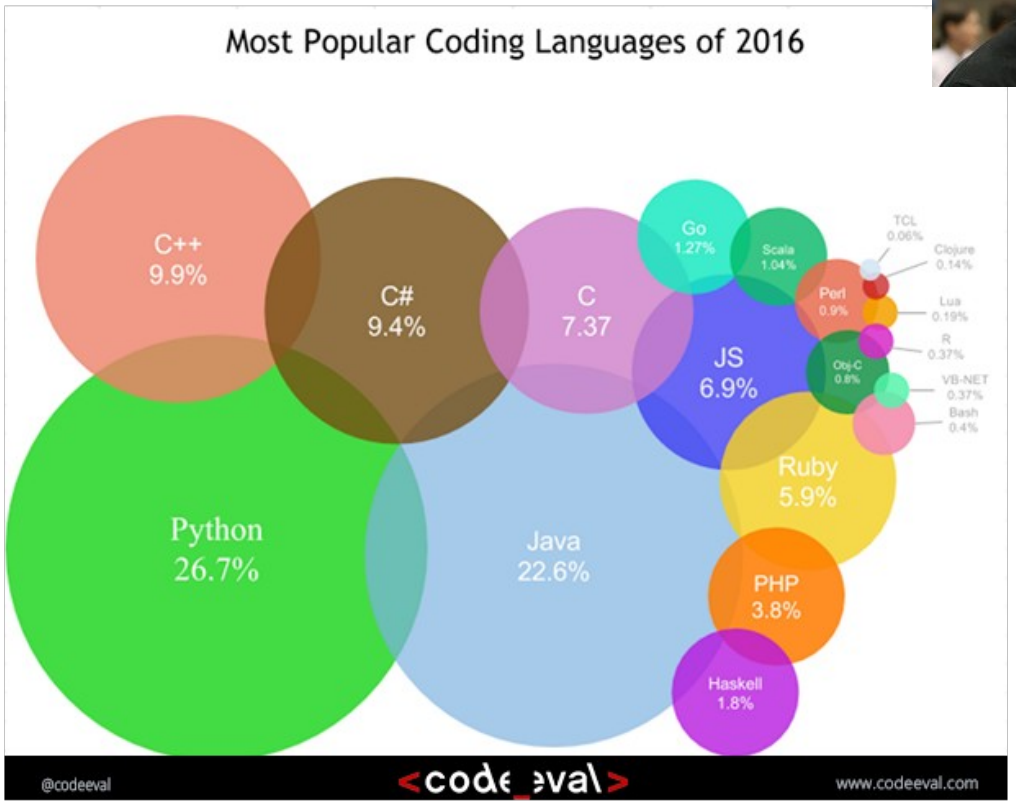

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Python 3: Fibonacci series up to n
>>> def fib(n):
>>>   a, b = 0, 1
>>>   while a < n:
>>>   print(a, end=' ')
>>>   a, b = b, a+b
>>>   print()
>>> fib(1000)
0 1 1 2 3 5 8 13 21 34 55 89 144 233 377 610
987
Functions Defined
The core of extensible programming is defining functions. Python allows mandatory and optional arguments, keyword arguments, and even arbitrary argument lists. More about defining functions in Python 3
1
2
3
4
5
Python is a programming language that lets you work quickly and integrate systems more effectively. >>> Learn More

Started by Guido Va Rossum as a hobby

Now widely spread

Open Source! Free!

Versatile





Guido Van Rossum by Doc Searls on Flickr CC-BY-SA

PYTHON TODAY

- Developed a large and active scientific computing and data analysis community
- Now one of the most important languages for
  - Data science
  - Machine learning
  - General software development



Many popular Python toolboxes/libraries:

- NumPy
- SciPy
- Pandas
- SciKit-Learn

Visualization libraries

- matplotlib
- Seaborn

All these libraries are installed on the SCC

and many more …

PYTHON LIBRARIES FOR DATA SCIENCE

NumPy: 

- introduces objects for **multidimensional arrays and matrices**, as well as functions that allow to easily perform advanced **mathematical and statistical operations** on those objects

- provides vectorization of mathematical operations on arrays and matrices which significantly improves the performance

- many other python libraries are built on NumPy

Link: http://www.numpy.org/

Python Libraries for Data Science

SciPy: 

- collection of algorithms for **linear algebra, differential equations, numerical integration, optimization, statistics** and more
- part of SciPy Stack
- built on NumPy

Link: https://www.scipy.org/scipylib/

Python Libraries for Data Science

Pandas: 

- adds data structures and tools designed to work **with table-like data** (similar to Series and Data Frames in R)
- provides tools for data manipulation: **reshaping, merging, sorting, slicing, aggregation** etc.
- allows **handling missing data**

Link: http://pandas.pydata.org/

matplotlib: matplotlib

- python 2D plotting library which produces publication **quality figures** in a variety of hardcopy formats
- a set of **functionalities similar to those of MATLAB**
- **line plots, scatter plots, barcharts, histograms, pie charts** etc.

Link: https://matplotlib.org/

SciKit-Learn:



- provides **machine learning algorithms: classification, regression, clustering, model validation** etc.

- built on NumPy, SciPy and matplotlib

Link: http://scikit-learn.org/

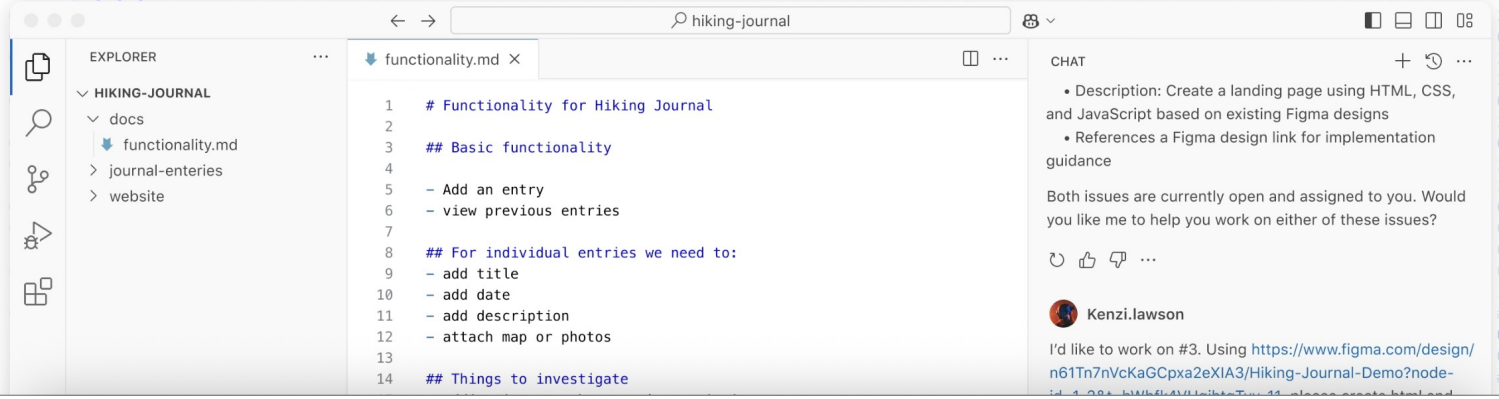
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The open source AI code editor

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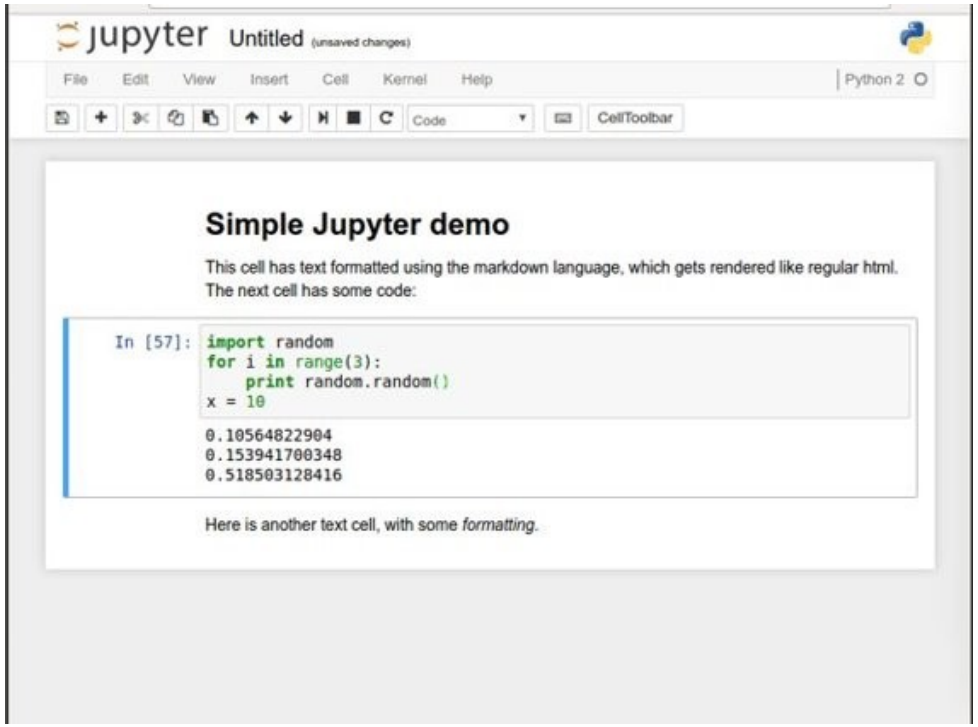
Web, Insiders edition, or other platforms

By using VS Code, you agree to its license and privacy statement.



NOTEABLE (JUPYTER NOTEBOOKS)

- Easy to use environment
- Web-based
- Combines both text and code into one
- Come with a great number of useful packages



Course Objectives

- Understand & prepare data
- Analyze and interpret data
- Apply and validate data science techniques

Learning Outcomes

- Analyze & process numerical and categorical structured data
- Apply data science methods to practical problems
- Cross-validate model outcomes
- Use programming tools to implement data science techniques

Course Topics

Intro to Python

- Data types: int, float, string, list, dict
- Control structures: if-else, loops
- Functions and modules
- Libraries: NumPy, Pandas, Matplotlib

Basic Programming Concepts

- Variables and operators
- Input/output operations
- Iteration and recursion
- Error handling & debugging

Course Topics

Basic Statistics

- Mean, Median, Mode
- Variance, Covariance, Correlation
- Confidence intervals
- Histogram & probability basics
- Implementation in Python

Data Understanding and Preparation

- Data acquisition and sources
- Cleaning and scrubbing data
- Handling missing values
- Feature selection and transformation
- Implementation in Python

Course Topics

Exploratory Data Analysis

- Descriptive statistics
- Data visualization techniques
- Detecting patterns & trends
- Identifying outliers
- Implementation in Python

Linear Regression Methods

- Simple vs. multiple regression
- Model evaluation
- Implementation in Python

Basic Classification Methods

- Bayesian classifier
- Decision trees
- k-Nearest Neighbours (k-NN)
- Implementation in Python

Basic Clustering Methods

- K-means clustering
- Hierarchical clustering
- Cluster evaluation
- Implementation in Python

Database, SQL & Big Data [Not Part of Exam]

- Relational databases: primary & foreign keys
- SQL basics: SELECT, JOIN, GROUP BY
- Introduction to Big Data

Assessment & Grading

- Homework: 5%
- Quizzes: 10%
- Project: 20% – Groups of 4–5
- Midterm: 20% (conceptual and coding)
- Final Exam: 40% (conceptual and coding)
- Class Participation: 5%

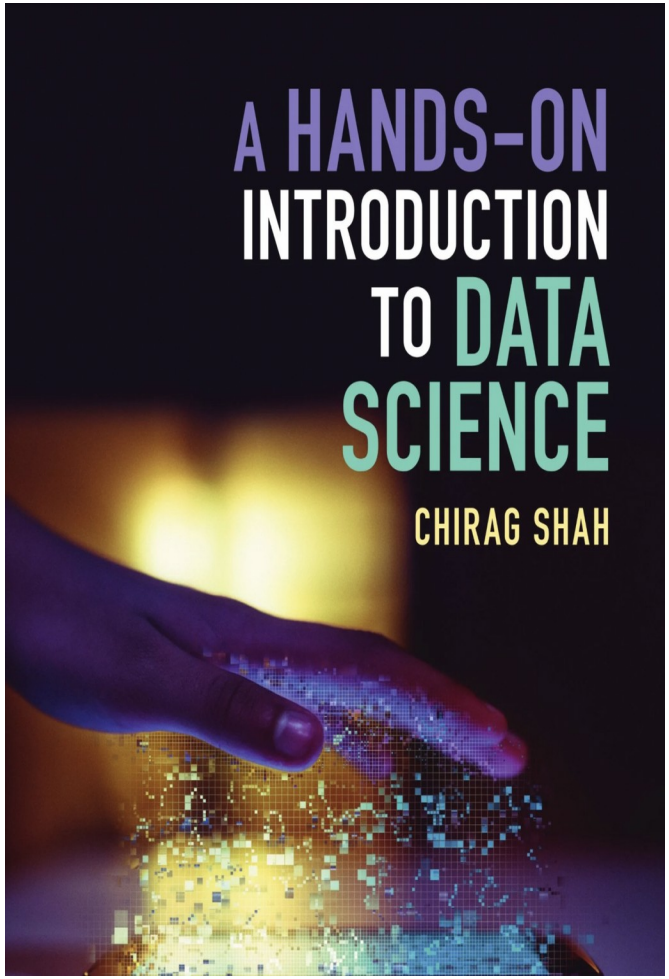
Student Responsibilities

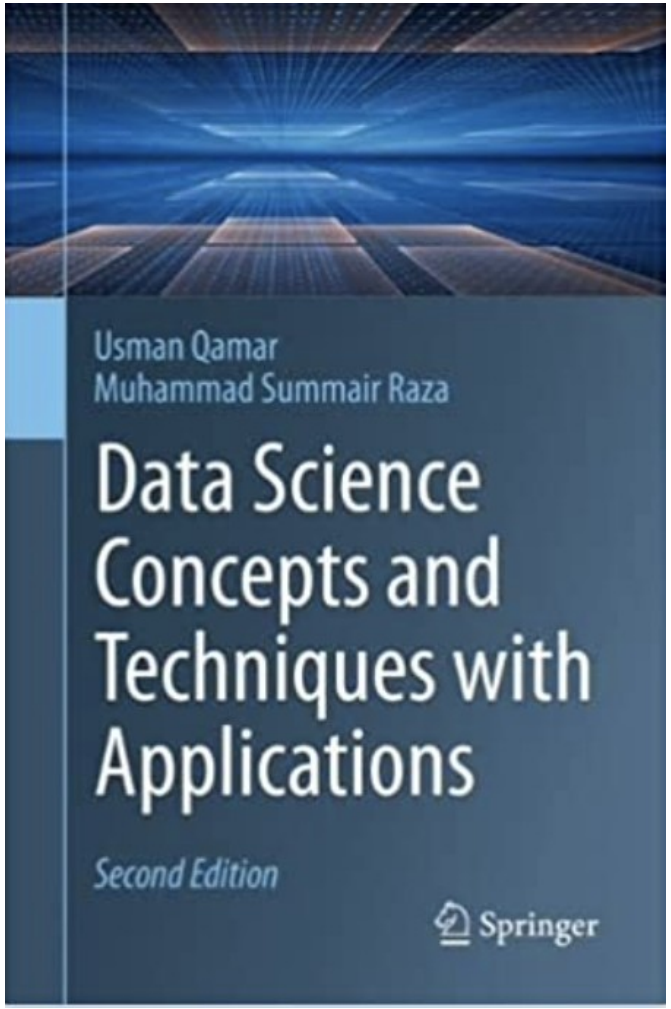
- Participate & engage in teamwork
- Keep up with lectures & deadlines
- Attend on time, stay until end of class
- Switch off mobile phones during class

Academic Integrity

- Cheating and plagiarism strictly prohibited
- Refer to KFUPM Guidelines for Conducting Exams

- **Recommended Textbooks:**





There is an inherent meaning in everything.

“Signs for people who can see.”